



Wrocław
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DNS Server

IT Infrastructure Management (W04IST-SI0826G)

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Spring 2026



This week

After this class, you should be able to:

- Explain how DNS resolves names using zones and resource records
- Distinguish between recursive and iterative DNS queries
- Describe the DNS server hierarchy and zone delegation
- Explain why Active Directory depends on DNS and which records it registers automatically

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Why this matters

- A new domain controller is promoted — clients still cannot log in
- `gpupdate /force` fails with a network error across the domain
- Users report they cannot reach the intranet site by name — but `ping` by IP works fine
- **The cause is almost always misconfigured or missing DNS — today you will learn how to find and fix it**

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Agenda

- 1 DNS Service
- 2 DNS and Active Directory

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Quick recall

Without looking at your notes:

- 1 What is an AD site, and why does it exist?
 - 2 Which FSMO role issues unique RIDs to domain controllers?
 - 3 What controls replication scheduling between sites?
-
- 1 A logical grouping of well-connected subnets — controls replication topology and DC locator
 - 2 The RID Master
 - 3 Site link cost and replication interval in **AD Sites and Services**

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What does a DNS server do?

- **Domain Name System (DNS)** — translates hostnames to IP addresses
- Without DNS, every network resource must be reached by IP address only
- Also enables **service discovery** — clients find DCs, LDAP, and Kerberos via DNS SRV records
- Every computer joined to a Windows domain *requires* a working DNS server

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DNS client configuration

- DNS client (e.g. a workstation or member server) needs:
 - Address of a primary DNS server (and optionally a secondary)
 - DNS suffix for the domain (e.g. contoso.com)
- In Windows: **Control Panel > Network > Adapter Properties > TCP/IPv4**
- Via PowerShell:

```
1 Set-DnsClientServerAddress -InterfaceAlias "Ethernet" `
2 -ServerAddresses "192.168.1.10", "192.168.1.11"
```

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Before we continue — predict

- A DNS server receives a query for server01.contoso.com
- It has no record for that name in its zone
- **What happens next — and who ultimately answers?**
- The resolver forwards the query or performs iterative resolution
- It contacts root servers → TLD servers → authoritative server for contoso.com
- The authoritative server returns the answer; the local resolver caches it

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What is a DNS zone?

- **Zone** — a portion of the DNS namespace managed by one authoritative server
- Stored as a file (traditional DNS) or in Active Directory (AD-integrated)
- Zone types:
 - **Primary** — the writable master copy; edited directly
 - **Secondary** — read-only copy; updated via *zone transfer* from the primary
 - **Stub** — contains only NS and A records pointing to the authoritative server
 - **Active Directory Integrated** — stored in AD, replicated automatically

Forward and reverse lookup zones

- **Forward lookup zone** — maps hostnames → IP addresses
 - Example: server01.contoso.com → 10.0.0.5
- **Reverse lookup zone** — maps IP addresses → hostnames
 - Example: 10.0.0.5 → server01.contoso.com
 - Named by reversed IP prefix: zone 0.0.10.in-addr.arpa
- Verify with:

1 `Get-DnsServerZone`

Zone resource record types

Record	Full name	Purpose
A	Address	Hostname → IPv4 address
AAAA	Quad-A	Hostname → IPv6 address
CNAME	Canonical Name	Alias for another hostname
NS	Name Server	Authoritative DNS servers for a zone
MX	Mail Exchange	Mail servers for the domain
SRV	Service	Location of services (LDAP, Kerberos)
PTR	Pointer	IP address → hostname (reverse lookup)

Recursive vs. iterative queries

- **Recursive query** — client asks its resolver to return a *final* answer
 - Used between workstation and its configured DNS server
 - Resolver does all the work; client gets an answer or an error
- **Iterative query** — DNS server asks another server, gets a *referral*, asks the next
 - Used between DNS servers during external resolution
 - Each server in the chain replies: "I don't know — try *that* server"



How DNS resolution works on the Internet

- Client sends a recursive query to the local DNS resolver
- Resolver queries **root servers** (13 addresses: a.root-servers.net - m.root-servers.net)
- Root server refers to the **TLD server** (e.g. .com, .pl)
- TLD server refers to the **authoritative server** for the domain
- Authoritative server returns the answer; resolver caches it until TTL expires

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Zone delegation

- **Zone delegation** — transferring authority for a sub-zone to a different DNS server
 - Example: contoso.com delegates emea.contoso.com to a server in the London office
- Implemented with **NS records** pointing to the delegated server
- Allows distributed management of large namespaces
- Required when creating child domains in an AD forest

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Admin challenge

After promoting a new DC, nslookup dc02.contoso.com returns “Non-existent domain” from workstations. The DC itself resolves the name correctly. As the admin, what are your first three diagnostic steps?

- 1 Check which DNS server workstations use: `ipconfig /all`
- 2 Verify the A record exists:
`Get-DnsServerResourceRecord -ZoneName contoso.com -Name dc02`
- 3 Confirm the AD-integrated zone has replicated to all DCs:
`Get-DnsServerZone` on each DC

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Why does AD depend on DNS?

- Active Directory components are discovered entirely through **SRV records**
 - Clients find a DC by querying: `_ldap._tcp.contoso.com`
 - Kerberos: `_kerberos._tcp.contoso.com`
- Without correct SRV records: logon fails, GPO processing fails, replication fails
- AD registers these records **automatically** when the DNS Server role is available

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The `_msdcs` domain

- When a domain is installed, two DNS zones are created automatically:
 - `contoso.com` — A records for all machines in the domain
 - `_msdcs.contoso.com` — records for **AD-specific services**
 - DC locator records (SRV, CNAME)
 - Global Catalog (GC) locator records
 - PDC Emulator locator records
- Separating `_msdcs` allows independent replication and delegation

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Zone replication in the domain

- AD-Integrated zones stored in the **Application partition** of the AD database
- Replicated automatically as part of AD replication — no manual zone transfer needed
- Replication scope options (**DNS Manager > Zone Properties > General**):
 - All DNS servers in the forest
 - All DNS servers in the domain (*default for `_msdcs`*)
 - All DCs in the domain
- No single point of failure — any DC can answer DNS queries authoritatively

<!-- ## In the lab this week

You will:

- Install the DNS Server role on Windows Server 2025
- Create a forward and reverse lookup zone
- Add A, CNAME, and PTR records manually
- Verify that AD-integrated zones exist and are populated after domain join

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Bibliography

- [1] O. Thomas, *Windows server inside out: Updated for windows server 2025*. Microsoft Press, 2025.
- [2] J. Krause, *Mastering windows server 2025 - fifth edition*. Packt Publishing, 2025.

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